

CA2 TSA Report

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1 Introduction

A personal motivation behind choosing this dataset and topic is an interest in gaining knowledge about markets and stocks.

In this project report, the relevant methodology referring to the project code will be discussed, highlighting key findings, model justifications, summarizing the workflow and thought process during the execution of this project.

The aim of this project will be the analysis of the closing price of Take-Two Interactive (ticker TTWO).

The project follows a structured workflow: data description and cleaning, exploratory data analysis, stationary testing, model selection, diagnostics, and forecasting. ARIMA(0,1,1) was selected as the final model following a grid search evaluated by AIC and BIC criteria, and a 10-step ahead forecast was generated beyond the last observed date.

2 Data Description

For the fetching of the data, the library `yfinance` is used. `yfinance` sources its data from Yahoo Finance and offers a reliable way for python users to download and access historical market data. One drawback of `yfinance` is that the stock availability depends on the volume of trading, in other words, stocks such as Greencoat Renewables PLC (Irish renewable energy company) where the stock has less trading volume has issues with its availability and the recorded data is not accurate or sometimes even readable.

Considering this drawback, a high volume stock must be considered. Therefore, series chosen for the implementation of this project's time series analysis is the stock price of Take-Two Interactive; TTWO is a video game publisher company known for publishing accomplished titles such as Grand Theft Auto and NBA2K. The company is considered a triple A company (AAA) which is the highest credit rating assigned by financial agencies, this in turn assures that the company has sufficient and reliable volume data to be analyzed.

The raw dataset was composed of Date, Open, High, Low, Close, Volume, Dividends and Stock Splits, as shown in the following table:

Column	Description	Type
Date	Format: dd/mm/YY, Frequency: Daily	Date
Open	Stock value when market opens	Decimal
High	Highest value recorded in a day	Decimal
Low	Lowest value recorded in a day	Decimal
Close	Stock value when market closes	Decimal
Volume	Number of shares traded in one day	Integer
Dividends	TTWO does not pay dividends	Boolean
Stock Splits	All values are zero — dropped	Boolean

Table 1: Dataset Variables Description

From those variables, Close price was chosen to be analyzed since it reflects most of the information accumulated during the trading session and is conventionally treated as a representative daily value. Close price helps reduce the impact of certain short-term noise such as overnight news and pre-market sentiment which are typically reflected by the Open price and may not reflect the overall trading session dynamics, furthermore High and Low often reflect intraday extremes rather than typical price levels.

Hence, Close is the ideal series for analysis in this context. (All other variables were dropped except Date)

2.1 Data cleaning

The date index was already in the correct datetime format and required no reformatting.

No missing values were identified, confirmed via the `.info()` output which details if there are missing values.

A box plot fig:1 was used for outlier detection and no outliers were identified in the Close price series.

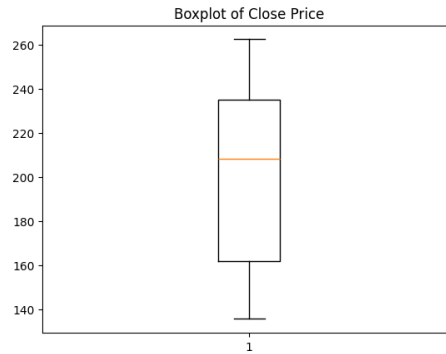


Figure 1: Box plot of Close price

3 Exploratory Data Analysis

In this section, it will be detailed how the dataset was visualized and the key features noted by these analysis.

3.1 Time Series Plot

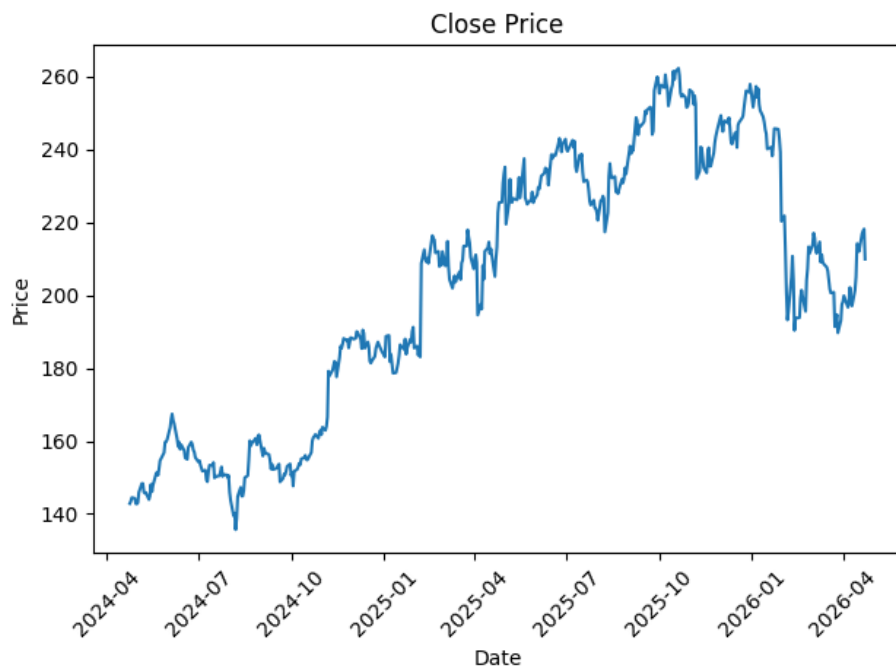


Figure 2: Time series plot

The price ranges from minimum \$140 to maximum \$260.

The series shows an uptrend from mid-2024 till mid-2025, with the price reaching its peak, followed by a price dip in February 2026 to the \$190 mark. This points to a significant change in market conditions or a major event in the company (such as earnings report). Visually the series showcases systematic mean change which tells us that it is not a stationary series.

3.2 Distribution

The distribution is not centered around a single price instead showcases 3 main price clusters.

TTWO shows a cluster of prices around the mode \$150, which indicates that the stock spent a significant amount of time trading at this lower range (during the earlier part of the sample).



Figure 3: Series distribution

Then, once the uptrend happened, the cluster around the mode \$240 price became noticeable.

The last cluster is around \$190 price which happened due to the February 2026 price dip and subsequent stabilization of the price.

In summary, the series showcases 3 clusters with 2 main modes and a interval range around \$190.

3.3 Lag Plot

Clearly there is a strong linear relationship between them, which translates to the yesterday values being very similar to today's values (e.g. if yesterday was high, today will also be high and vice-versa).

3.4 Autocorrelation

By observing the plot, recent lags ($\text{lag} = 1, 2, \dots, 10$) show strong positive autocorrelation, while by around lag 150 the correlation coefficient is essentially null relative to the current time step. This means that the link between current and past values becomes weaker as the lag increases.

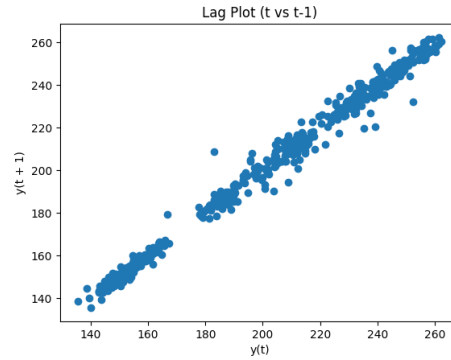


Figure 4: Lag plot

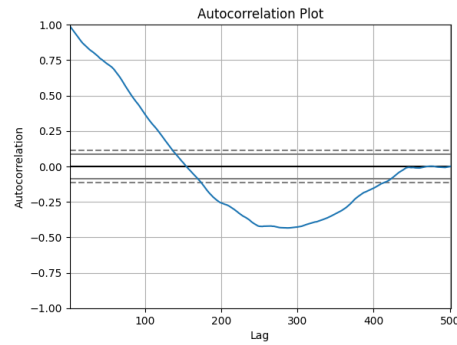


Figure 5: Autocorrelation plot

A very weak negative correlation appears around lag 250, suggesting that price movements more than half a year apart show a slight inverse relationship. However, this is clearly caused by the sharp price drop in February, which affects the autocorrelation pattern. Therefore, this observation does not provide any meaningful conclusion.

3.5 Decomposition

There are two main ways of performing decomposition:

- Classical decomposition
- STL decomposition

The classical decomposition method separates the series into trend, seasonal, and residual components; it assumes fixed, repeating seasonality. The trend is estimated using global moving averages. Seasonality needs to be known and

constant, which in the case of TTWO is not so clear since most price moves are not based of a seasonal trend as seen during the last steps.

STL decomposition is used to separate the series into trend, seasonal, and residual components. STL is more robust to outliers, works with seasonal periods and can change over time (improvement from the global moving averages). The trend and seasonality components are estimated using LOESS smoothing.

The reasoning behind choosing STL instead of the classical (via the method `seasonal_decompose`) decomposition is due to the lack of periodic events in the TTWO price, which reflects a lack of seasonality. There are trends, volatility clusters, white noise, and price shocks; however, seasonality is hard to justify. Therefore, applying global moving averages to decompose the series would not be as precise as using local weighted regression, which is more flexible and resistant to noise.

Both output plots were quite similar visually and logically; however, STL does show smoother trends. Looking at the residual plot, in the classical decomposition the residuals extend to around -20 , meaning the model's fitted value was about \$20 higher than the actual price. Visually the graph seems smoother, but on the same scale there is clearly clustering around February during the price drop. On the contrary, the STL residual graph looks like white noise (with a similar nuance during February, but only around -10 , meaning the fitted value was about \$10 lower than the actual price), as it is supposed to.

Note that the "season" plot (third plot) illustrates that the patterns are not repeating or constant over the time axis, there are some local patterns occurring however they break after a while and do not repeat ever again, therefore there is no evidence of seasonality.

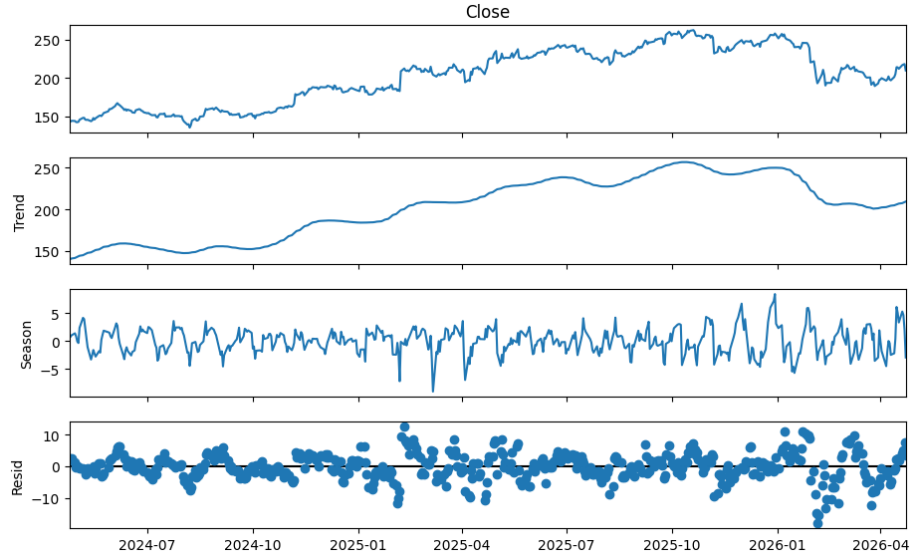


Figure 6: STL decomposition

4 Methodology

4.1 Stationarity

Before modeling, it is necessary to confirm whether the series is stationary. Two standard stationarity tests were applied:

- ADF test: null hypothesis is that the series is non-stationary
- KPSS test: null hypothesis is that the series is stationary

The ADF hypothesis test tells us that the series is not stationary. The test result did not provide enough evidence to reject the null hypothesis. In other words, the series behaves like a random walk, with changing mean and variance over time.

The KPSS test rejects the null hypothesis of stationarity (stat = 3.03, $p = 0.01$). The series is non-stationary.

Both the ADF and KPSS outcomes are consistent with the visual inspection, which already indicated non-stationarity due to shifting mean values and changes in the spread of values.

4.2 Log-Differencing

The simplest method to detrend a time series is by differencing. Log-differencing was applied to the Close price series to stabilize both the mean and variance.

By observing the transformed series, the mean of the differenced series is close to 0. There is no clear upward or downward trend, and it looks like white noise (normally distributed, based on the histogram). The variance also looks constant, meaning the trend component has been removed by performing log-differencing.

An ADF test on the differenced series confirmed stationarity.

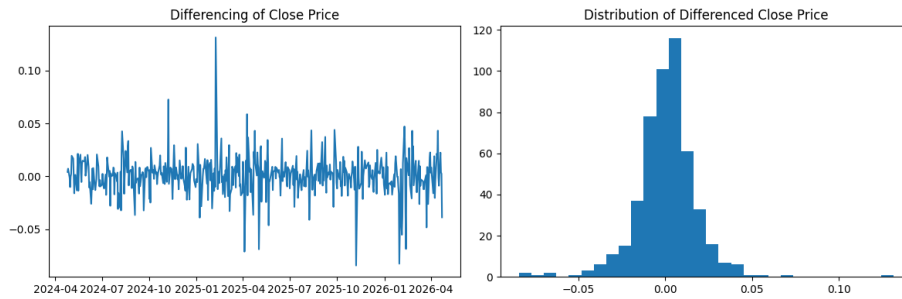


Figure 7: Log Differenced Close Price

4.3 ACF and PACF

ACF (Autocorrelation Function) measures the correlation between a time series and its own lagged values, helping to identify overall trends and seasonality in the data. This includes all lag effects mixed together. ACF helps determine the MA structure.

PACF (Partial Autocorrelation Function) measures the partial correlation of a stationary time series with its own lagged values, controlling for the values of the time series at all shorter lags. PACF helps determine the AR structure.

The ACF and PACF of the log-differenced series show a single significant spike at lag 1, with all other lags inside the confidence bands.

In general, when there are no spikes it suggests that the differenced series is composed of white noise, which is typical of efficient markets.

This indicates that only short-term dependence remains, supporting the use of simple ARIMA models such as:

- ARIMA(1,1,0)
- ARIMA(0,1,1)
- ARIMA(1,1,1)

5 Model Selection

5.1 Train-Test Split

The last 30 trading days are used as the test set. This keeps the evaluation simple and gives a clear forecast accuracy section.

5.2 Naive Baseline

Before fitting ARIMA, a persistence model is used as a simple baseline. This predicts that the next close price will be the same as the previous one, following the high correlation value from the previous lag relative to the current lag.

Note that the naive model is mathematically equivalent to an ARIMA(0,1,0) process and serves as the standard benchmark for financial time series.

5.3 ARIMA

ARIMA (Auto Regressive Integrated Moving Average) is the preferred model over something like SARIMA (Seasonal-ARIMA) because in this data there is no seasonality component. After log-differencing there is no clear trend and variance is roughly constant, after performing the ADF test on the differenced series it is clear that the series can be considered stationary.

From the three candidate models, ARIMA(0,1,1) has the lowest AIC and BIC values, while ARIMA(1,1,0) produces nearly identical results. ARIMA(1,1,1) performed worse due to the additional parameter.

Considering the results from AIC and BIC alone ARIMA(0,1,1) would be the wisest choice. These results were to be expected since their ACF and PACF plots were very similar to begin with.

In broader financial terms, even though past values are relevant (where AR(1) is more prevalent), stocks and markets react to unexpected news and not only past values, earnings surprises, geopolitical tensions, macro market announcements such as monetary policy, etc. Therefore MA(1) is a more prevalent model considering this prior knowledge.

Order	AIC	BIC
(0,1,1)	2586.941349	2595.246815
(1,1,0)	2586.941721	2595.247186
(1,1,1)	2587.985775	2600.443973

Table 2: Candidate ARIMA Models AIC and BIC

5.4 Grid Search

A grid search was performed over a range of p, d, q values evaluated by AIC and BIC.

Order	AIC	BIC
(2,1,3)	2581.569735	2606.486131
(0,1,0)	2584.944005	2589.096737
(1,1,3)	2585.108714	2605.872377
(3,1,1)	2585.177399	2605.941063
(3,1,0)	2586.123623	2602.734554
(0,1,3)	2586.401601	2603.012532
(2,1,0)	2586.666197	2599.124395
(3,1,2)	2586.757229	2611.673625
(0,1,2)	2586.904439	2599.362638
(0,1,1)	2586.941349	2595.246815

Table 3: Grid Search Results sorted by AIC (Top 10)

In terms of AIC, ARIMA(0,1,1) actually performs the worst compared to the other candidate models, however their AIC values are very close nonetheless. ARIMA(2,1,3) has too many parameters so it is likely to overfit the data, hence it would probably not be a good choice.

Examining the best models based on BIC, ARIMA(0,1,1) performs the best after the base model.

5.5 Final Model Selection

Based on the AIC and BIC values, ARIMA(0,1,1) aside from the baseline is the model with less parameters and has the lowest BIC. Recalling that BIC penal-

izes model complexity and that AIC does not penalize overfitting sufficiently, ARIMA(0,1,1) is arguably the best model over the rest of the candidates. Since the final goal is out-of-sample prediction, BIC is the more appropriate criterion here, making the higher AIC of ARIMA(0,1,1) a secondary concern.

No model simultaneously minimizes both AIC and BIC, so committing to one criterion based on the modeling objective is necessary. ARIMA(0,1,1) has the best statistics out of the group.

Why not choose just the base model? The base model although it scores relatively well compared to the other models, is not entirely suitable since it is merely capturing the fact that the previous lag prices are highly correlated with the current prices, this is already known from the ACF and PACF plots. The model does not bring any new information or prediction value beyond a naive lag structure, compared to the other models that at least deviate from that known fact.

Final Selection: ARIMA(0,1,1)

6 Diagnostics

After fitting ARIMA(0,1,1), the diagnostics section will be used to analyze residuals. A well designed model should make residuals behave like white noise, uncorrelated, centered at 0 and normally distributed.

6.1 Residuals Over Time & Distribution

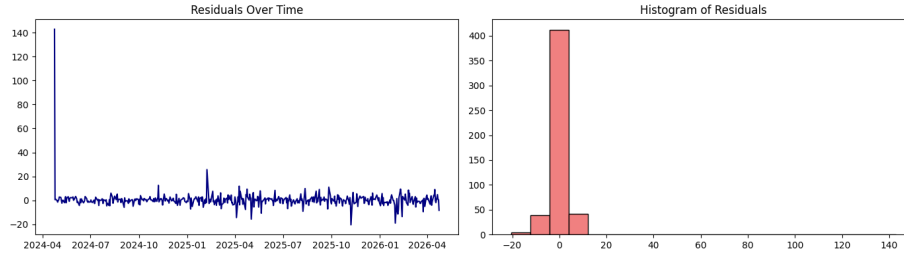


Figure 8: Residuals over time

The residuals for the majority of the series are centered at zero, which is consistent with white noise behavior. The large spike at the first residual is due to the fact that the initial prediction is based solely on the intercept; once the MA(1) correction kicks in from the second observation onward, the residuals center around zero with no systematic drift or pattern.

6.2 Q-Q Plot

The residuals approximate normality (with the first value exception discussed).

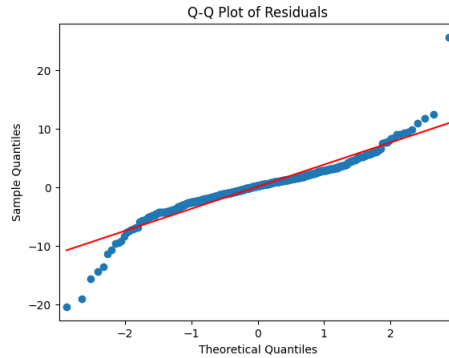


Figure 9: QQ-residuals

There are deviations at both tails which tells that the distribution has heavy tails making a mild "S" shape. It means that extreme price movements occur more frequently than a normal distribution would predict, which is expected behavior in financial time series since prices are traded by headline news which might spike the price unexpectedly.

6.3 Ljung-Box Test

The Ljung-Box test tests whether the residuals are independently distributed up to lag 10.

The null hypothesis is that there is no autocorrelation in the residuals. A p-value above 0.05 means we fail to reject the null, which is the desired outcome and confirms the model has captured the dependence structure of the series.

$0.468356 > 0.05 \rightarrow$ fail to reject null hypothesis, no autocorrelation in the residuals.

Hence, ARIMA(0,1,1) has successfully captured the dependence structure of the series and the residuals behave as white noise.

7 Forecasting

With the model validated through diagnostics, forecasting is performed in two stages:

- Forecasting against the test data
- Future forecasting

First, the model trained on the training set is evaluated against the test data to quantify predictive accuracy. Second, the model is refitted on the full dataset and used to generate a genuine 10-step ahead forecast into the future.

7.1 Test Forecast

The ARIMA(0,1,1) forecast over the test period produces a flat forecast around \$210. Since ARIMA(0,1,1) has no auto-regressive terms, the forecast converges to a constant after the first step.

The actual test data however shows continued volatility, dipping to around \$190 before recovering above \$210. The 95% confidence interval is wide and does contain the actual values for most of the test period, meaning the model is not wrong, but it is not tracking the short term movements either.

This is expected behavior for a financial series where price changes are largely unpredictable beyond the immediate next step. In an efficient market scenario, stock prices are modeled as stochastic processes which incorporate a random component that cannot be deterministically predicted.

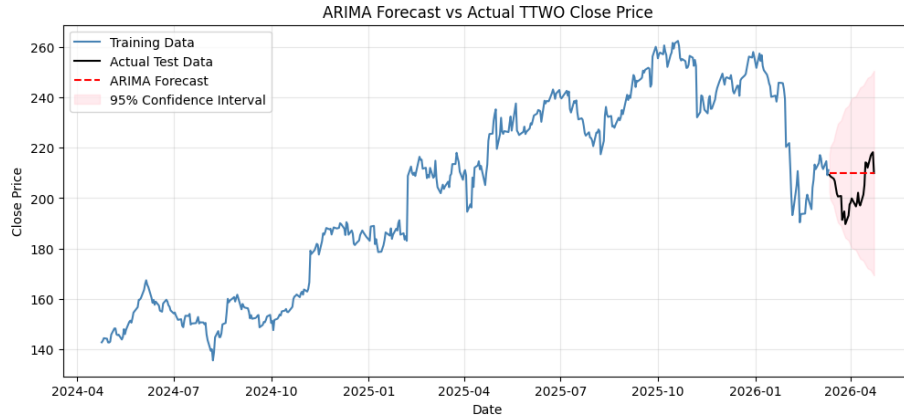


Figure 10: Test Forecast

7.2 Accuracy Metrics

MAPE (Mean Absolute Percentage Error) expresses forecast error as a percentage of the actual values, making it easier to interpret in relative terms regardless of the price scale.

ARIMA(0,1,1) was compared against the baseline model ARIMA(0,1,0) across three error metrics: RMSE, MAE, and MAPE. Lower values indicate better predictive accuracy.

Model	RMSE	MAE	MAPE
Baseline	3.785035	2.866999	1.418605
ARIMA(0,1,1)	10.350241	8.848155	4.443202

Table 4: Forecast Accuracy Metrics — Baseline vs ARIMA(0,1,1)

Unsurprisingly, the baseline outperforms ARIMA(0,1,1), which is a consequence of the baseline updating its prediction based on the actual last observed price at every step, giving it very high accuracy.

However, in a longer term context the baseline would probably not generalize well since its accuracy depends entirely on the most recent lag.

7.3 Future Forecast

After validating the ARIMA(0,1,1) model with the train and test sets, a 10-step forecast was generated; 10 business days beyond the last observed date. The model was refitted on the full dataset (train + test) to make use of all available information.

The widening confidence interval reflects the compounding uncertainty of multi-step forecasting, which is a representation of the limits of stock price prediction.

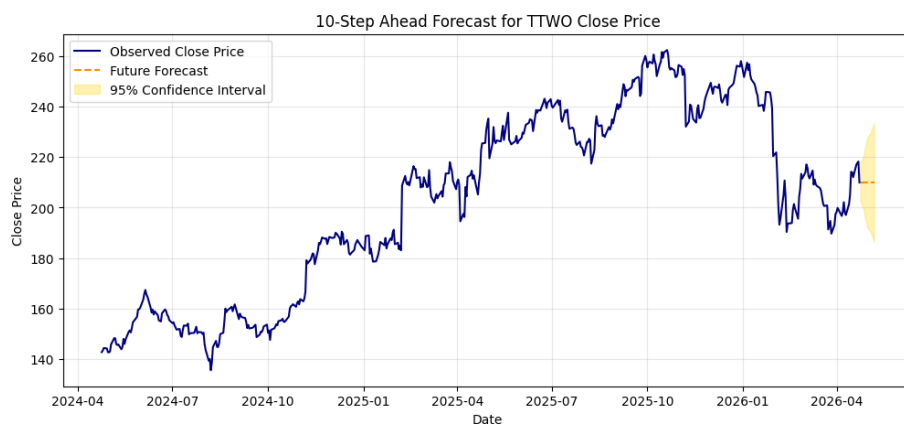


Figure 11: Future Forecast

8 Conclusion

This project analyzed and forecasted the closing price of Take-Two Interactive (TTWO) stock over a two-year period from April 2024 to April 2026.

The exploratory data analysis confirmed that the series is non-stationary, clearly showing an upward trend throughout 2025, followed by a sharp price drop in February 2026.

The ADF and KPSS tests corroborated the fact that the series is non-stationary.

Log-differencing was applied to achieve stationarity, producing a series that resembles white noise with roughly constant variance.

The ACF and PACF of the differenced series indicated short-term dependence at lag 1 only, which motivates the use of simple ARIMA models.

Following a grid search evaluated by AIC and BIC, ARIMA(0,1,1) was selected as the final model. While it did not achieve the lowest AIC score, it stood out with the lowest BIC score following the baseline, which is important since the objective is out-of-sample forecasting, choosing a model that is less complicated while generalizing well without overfitting is ideal.

The MA(1) structure from ARIMA(0,1,1) is consistent with prior financial knowledge where prices react to unexpected events, such as news or earnings reports, rather than following a pure auto-regressive process.

During the diagnostics section it was confirmed that the model generalized well, with the residuals resembling white noise while being approximately normally distributed. The Ljung-Box test corroborates this affirmation.

In terms of forecast accuracy on the test set, ARIMA(0,1,1) was compared against the baseline model ARIMA(0,1,0). The baseline outperformed ARIMA(0,1,1) since it directly depends on the previous lag at every step, achieving very high accuracy comparatively, however in a longer term period it is likely it would not perform well considering how stocks react to market events.

Finally, a 10-step ahead forecast was generated using the full dataset, providing a flat forecast (since ARIMA(0,1,1) has no auto-regressive term) bounded by the 95% confidence interval, reflecting the range within which the price is most likely to trade over the next 10 business days.

8.1 Limitations

Data limitations, this report only considered the past two years of data; seasonal trends may be present if a longer time period is taken into account.

Only one stock was evaluated, hence the model would most likely not generalize well versus other stocks.

ARIMA assumes linearity, sudden price drops such as the February 2026 crash are nonlinear events that a linear model may not fully capture.

Flat multi-step forecast, since ARIMA(0,1,1) lacks an auto-regressive structure, its usefulness is limited over longer time horizons beyond the first step.

Only one variable was modeled (Close price), external variables such as trading volume, market index movements, and sentiment analysis would complement the model and potentially improve forecast accuracy.

8.2 Future Work

Future work regarding this project would involve exploring alternative modeling approaches, particularly machine learning models:

- Regression models
- FNN (Feedforward Neural Network)
- CNN (Convolutional Neural Network)

- LSTM (Long Short-Term Memory)

Followed by a systematic comparison of their forecast accuracy against ARIMA(0,1,1) using consistent evaluation metrics.

Additionally, alternative statistical models such as Prophet were not explored, which could be another alternative model technique to further compare alongside ARIMA(0,1,1).